

Neural correlates of successful costly punishment in the Ultimatum game on a trial-by-trial basis

Abbreviated title: Costly punishment in the Ultimatum game

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Supplemental Online Material

A Reference Task

We used a reinforcement learning task as a reference task to measure neural responses to facial stimuli in a non-costly-punishment setting. Comparison of the neural responses in this game with those in the Ultimatum game allows for disentangling the effects that are specific in the context of the Ultimatum game (i.e., costly punishment).

The task of the participant is to offer each of four partners (2 male, 2 female) a fruit that he or she likes, as indicated by a smiling facial expression of the partner, and to avoid offering fruits that he or she does not like, as indicated by a neutral or sad facial expression. On each trial, the participant sees a picture of the partner with a neutral facial expression (see figure S1). Next, five different fruits are shown. The participant chooses one of them. Next, the facial expression of the partner (smile, neutral, sad) signals how she or he likes the chosen fruit. Thereby, only one fruit is mainly associated

with a smiling facial expression (80%, and 10% for sad and neutral facial expression, respectively); all other fruits are mainly associated with either a neutral or sad facial expression (40% each, and 20% smiling facial expression). In total, there are 120 trials (appr. 5 minutes), 30 with each of the four partners. We investigate neural responses (N170, FRN, P3) following the facial expression.

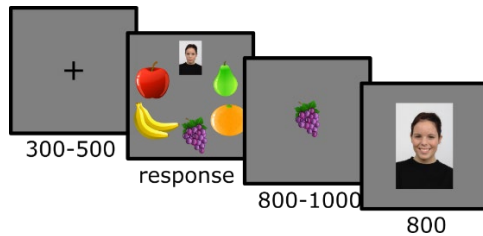


Figure S1: Timeline of the reference task

We investigate neural responses (N170, FRN, P3) following the facial expression, see Figure S2. Pre-processing was analogous to the main task. We quantified the N170 as the mean amplitude in the interval ± 20 ms around the negative peak in time window between 140 and 176 ms according to the grand average at electrodes TP9 and TP10 (as in the main task). The voltage at the two electrodes was subsequently averaged. The FRN was quantified in time window between 216 and 292 ms at electrode FCz (i.e., the mean of the time window is 30 ms earlier as compared to the main task, according to the peak of the N2 component in the reference task). The P3 was quantified in the time window between 456 and 588 ms at electrode Pz (as in the main task).

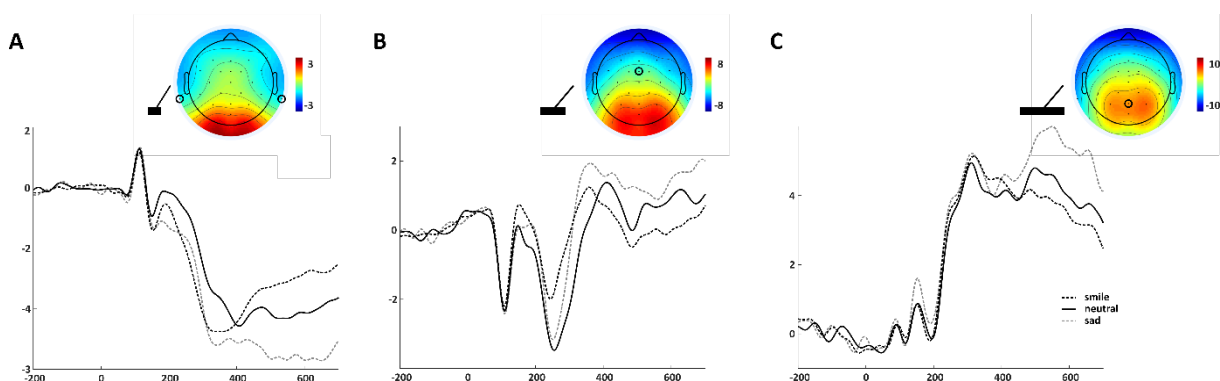


Figure S2: ERP following the presentation of the emotional facial stimulus in the reference task. (A)

N170 at TP9/TP10, topoplot averaged in the time range from 140 to 176 ms according to the grand

average. (B) FRN at FCz, topoplot averaged in the time range from 216 to 292 ms. (C) P3 at Pz, topoplot averaged in the time range from 456 to 588 ms.

The average number of trials per condition for the emotional face was 61, 28 and 29 for the smiling, neutral, and sad facial expression, respectively.

Results

Participants offered the correct fruit in 54% of the trials. Confidence intervals [.48 .59] indicate that this value was above guess probability of 20%. A generalized linear mixed model revealed a positive influence of (standardized) trial number on correct responses ($F=301$, $p<.001$, estimate .48). As can be seen in Figure S3, rate of correct responses started around guessing probability and rose asymptotically toward a rate of .62.

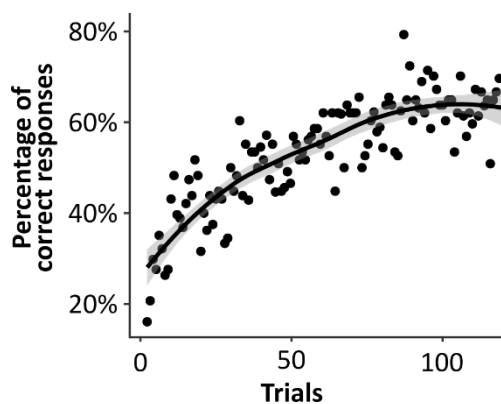


Figure S3: Average number of correct responses across participants according to trial number; shaded area shows the standard error around the predicted values.

Psychophysiological reactions on the presentation of the face were analyzed with linear mixed model with the fixed effects face (smiling, neutral, and sad facial expression) and (standardized) trial number and random intercept for participants. For N170, a significant effect for face was found ($F=5.8$, $p=.003$). Fixed effects revealed an arousal effect, with more negative amplitudes for both smiling ($-1.13 \mu V$, $p=.002$) and neutral ($-1.16 \mu V$, $p=.003$) facial expressions, compared to neutral ($-0.62 \mu V$).

For the FRN, we found a significant effect for face ($F=30.1$, $p<.001$). Fixed effect revealed more positive amplitudes for smiling compared to sad facial expressions ($p<.001$) and smiling compared to neutral facial expression ($p<.001$). Additionally, sad facial expressions were also more positive compared to neutral ($p=.009$). There was also a significant effect of trial number ($F=18.2$, $p<.001$) and a significant interaction between trial number and face ($F=3.6$, $p=.03$). Amplitudes increased and differences between conditions decreased over time as learning took place.

For the P3, a significant effect for face was found ($F=12.4$, $p=.006$). Sad facial expressions elicited more positive amplitudes, compared to smiling ($p<.001$), but also compared to neutral facial expressions ($p=.006$).

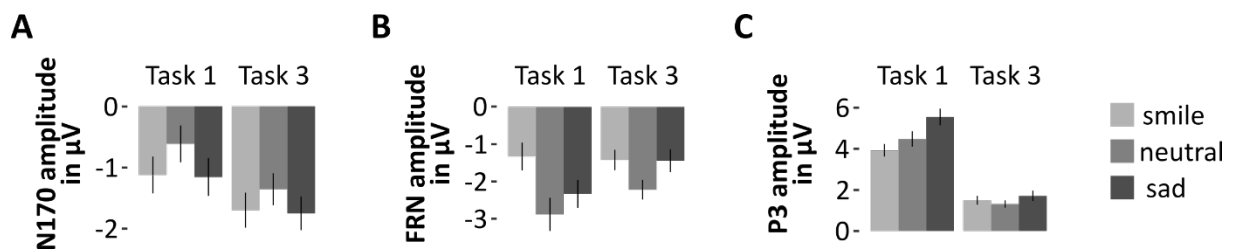


Figure S4: Neural responses to facial expressions in the references task (fruit game; Task 1) and the main task (Ultimatum game as responder, Task 3) for N170, FRN, and P3. Error bars indicate standard error of the mean.

We also compared the effect of facial expression on neural responses in the reference task (fruit game, Task 1) to those in the main task (Ultimatum game as responder, Task 3). For direct comparison, we coded the two conditions of neutral facial expression after rejection and neutral facial expression after acceptance in Task 3 both as neutral. We repeated the analyses, as described above, and included an additional fixed effect for *task* with two levels, Task 1 and Task 3. Descriptive results are depicted in Figure S4.

For the N170, we found a main effect for task ($F = 45.9$; $p < .001$), indicating more negative amplitudes in Task 3, compared to Task 1. Importantly, the interaction between task and facial was not significant ($F = 1.5$; $p = .22$), indicating that the effect of facial expression on N170 amplitudes was not different

in Task 1 and Task 3. As noted above, responses in both tasks follow an arousal effect, with more negative amplitudes for emotional, compared to neutral facial expressions.

We found an interaction between task and facial expression for FRN amplitudes ($F = 11.2$, $p < .001$). In line with the reversed FRN effect, in Task 3, sad facial expressions elicited a feedback positivity similar to smiling facial expressions. There was no significant difference between these two expressions ($p = .81$). Contrary, sad facial expressions showed a stronger negativity compared to smiling facial expressions in Task 1, see above ($p < .001$). We also found a main effect for task, indicating overall more negative amplitudes in Task 1 compared to Task 3, maybe due to the number of conditions that were perceived as positive versus negative feedback in the context of the two tasks.

For the P3, there was a huge main effect for task ($F = 450$, $p < .001$) which was qualified by an interaction between task and facial expression ($F = 8.7$, $p < .001$). The pattern of results suggests that P3 amplitudes were larger for Task 1, compared to Task 3, and that the effect of facial expression was larger in Task 1, compared to Task 3, presumably because information from facial expressions were task-relevant in Task 1 (participants inferred from facial expression whether the fruit was liked by the participant or not, which was the instructed task). Contrary, in Task 3, participants could, but did not have to (and were not instructed to) use the information from facial expressions for upcoming trials.

Discussion

The purpose of the reference task was to estimate neural responses to emotional facial expressions in a non-costly punishment context within the same sample as the main task. We found an arousal effect for the N170, with stronger negativity for emotional facial expressions (smile, sad) compared to neutral. In addition, we found a valence effect for the FRN, with a reward positivity for smiling facial expressions compared to neutral and sad; and a motivational effect for the P3, with stronger positivity for sad (but not neutral) compared to smiling facial expressions indicating a need for behavioral adjustment.

The results of the N170 are in line with a broad literature showing that the N170 is not only sensitive for facial compared to non-facial stimuli, but also to emotional compared to non-emotional facial expressions (Blau & McCandliss, 2007; Eimer, 2000; Hinojosa et al., 2015; Pizzagalli et al., 2002; Rossion et al., 2000). Comparisons between the references task and the main task (Ultimatum game, see main manuscript) revealed a similar pattern in both tasks. Thus, the context of the two games did not modulate the early perception of facial expressions.

Results for the FRN are also in line with literature regarding the valence effect of positive compared to negative stimuli eliciting a reward positivity. Against the backdrop of reinforcement sensitivity theory (Sutton & Barto, 1998), events that are better than expected elicit a positive temporal difference error whereas events that are worse than expected elicit a negative temporal difference error which allows for learning from experience and, ultimately, adjusting behavior (Holroyd & Coles, 2002; Miltner et al., 1997). Within the present paradigm, more negative amplitudes for the sad and neutral facial expression are also in line with the theory of surprise (Alexander & Brown, 2011) underlying the FRN component for trials in which participants had expected a smiling facial expression (after they formed a hypotheses on the correct response, and especially for trials in which they pressed the correct button but received negative feedback). In addition to our expected hypotheses, we also found a small difference between neutral and sad facial expressions. Neutral facial expressions are rather ambiguous and, thus, more difficult to interpret. In addition, the sad facial expressions might have been a better match to the task ("I do not like the fruit") than the neutral expression, which intuitively suggests a more indifferent reaction. These effects might have required additional cognitive resources and thus a more pronounced FRN. Importantly, the comparison with Task 3 showed a remarkably different pattern of facial expressions in the context of the Ultimatum game. As discussed in more detail in the main manuscript, this difference is due to the reversed FRN in the context of successful punishment. Here, the sad facial expression is the desired outcome that signals that costly punishment was successful and, thus, elicits a feedback positivity similar to the smiling facial expression after acceptance.

The P3 also shows an expected pattern with stronger positivity in trials which require behavioral adjustment (indicating that the wrong fruit was chosen, and the cognitive representation requires adjustment). There was again an unexpected difference between the neutral and sad facial expressions, with stronger positivity for the latter. As for the FRN, it might be speculated that the sad facial expression more intuitively suggests a need for behavioral expression, compared to the neutral facial expression, and thus elicits a larger P3. The large differences between P3 responses in Task 1, compared to the main Task 3 (Ultimatum game) can be explained due to differences in the task relevance of the facial expressions: Whereas participants were instructed to infer task-relevant information from the facial expressions in Task 1 (i.e., does my partner like the fruit?), this was not the case in Task 3. In the latter, participants may have incorporated the information for upcoming decisions on a voluntary basis, despite not being instructed to do so.

A limitation of the newly developed paradigm is that the number of outcomes (like vs. dislike the fruit) does not match the number of emotional facial expressions we used (smile, neutral, sad). In retrospect, some of the unexpected findings might be due to the assignment of both sad and neutral facial expressions to the same experimental condition.

In sum, we were able to replicate key findings regarding the neurophysiological responses to emotional facial expressions within our sample. Taking the above mentioned limitations into account, the results can provide more direct evidence to effects that are (or are not) specific to the costly-punishment context of the main task, compared to the non-costly punishment context of the reference task.

B Supplemental analyses

As suggested by an anonymous reviewer, we investigated whether neural responses to the presentation of facial expressions might be influenced by expectations. For example, participants might expect a fair offer because they made fair offers themselves or because they just rejected an offer to penalize the proposer. Such expectancies might influence how emotional facial expressions are perceived and processed. As an indicator of expectancies during a trial, we investigated the neural response to the presentation of the offer. In particular, we investigated FRN and P3 amplitudes. FRN amplitudes might indicate whether an offer was better or worse than expected. P3 amplitudes might be an indicator of the salience and motivational relevance of a stimulus and might thus be associated with the current engagement during a trial.

Methods.

EEG-data were preprocessed analogously to the presentation of the facial expressions, see method section of the main study. According to peak analysis and visual inspection of the event related potential (see Figure S5), we quantified the FRN as the mean amplitude in the interval between 288 and 372 ms. The P3 was quantified as the mean amplitude in the interval between 432 and 540 ms. N170 amplitudes were not considered as these are specific for the processing of faces and, thus, not applicable for the presentation of the pie chart symbolising the offer.

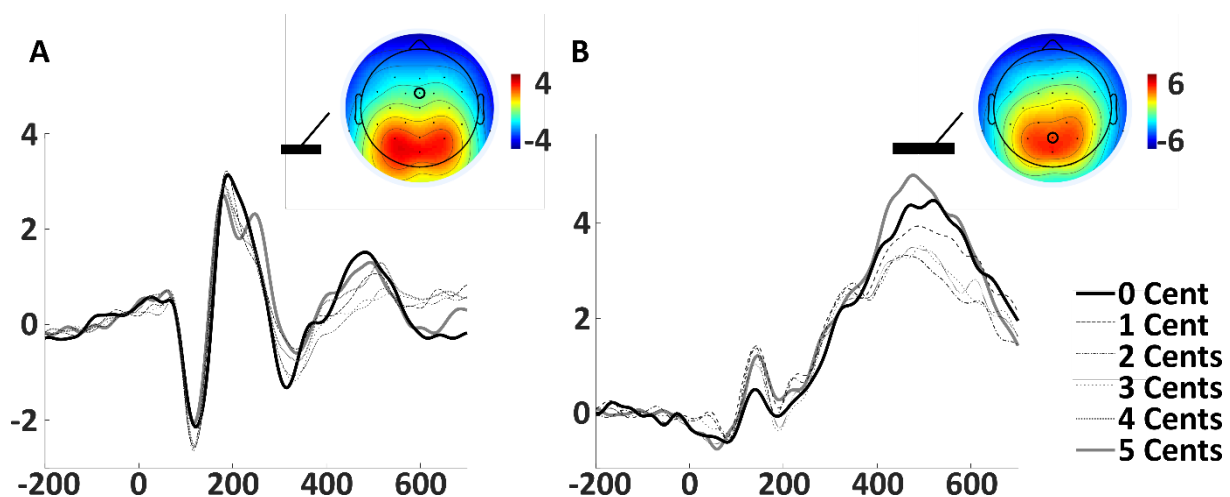


Figure S5: ERP following the presentation of the offer. (A) FRN at FCz, topoplot averaged in the time range from 288 to 372 ms. (C) P3 at Pz, topoplot averaged in the time range from 432 to 540 ms.

Statistical analyses

We extended the analyses on psychophysiological responses (N170, FRN, and P3) to the facial feedback, as described in the main manuscript. Therefore, to the linear mixed-effects model with the fixed effects *decision of the participant* (accept vs. reject) and *facial feedback* (more positive, i.e., smile after accept or neutral after reject, compared to more negative, i.e., neutral after accept and sad after reject) and random intercept for *participants*, we added an additional fixed effect for the z-standardized neural response (FRN and P3, in separate analyses) to the offer.

Results

First, for the N170 following the facial expression, we found neither an effect for the FRN nor for the P3 following the offer. Second, for the FRN following the facial expression, we found a moderating effect of the FRN following the offer on the interaction of decision and facial feedback (which characterizes the reversed FRN). In particular, the three-way interaction between decision, facial expression, and FRN amplitudes following the presentation of the offer on FRN amplitudes following the facial expression was $F = 8.6$, $p < .01$. As depicted in Figure S6A, the reversed FRN-effect was more pronounced when FRN amplitudes following the offer were more positive, compared to more negative. Post-hoc tests showed that the simple effect for FRN following the offer was significant for more negative facial expressions (i.e., sad) following rejection (i.e., successful punishment) and for more positive facial expressions (i.e., smiling) following acceptance (i.e., honored reward).

As FRN amplitudes are strongly affected by the valence of a stimulus, we also looked at these two effects separately for the 6 offers. It turned out that the effect of FRN amplitudes following the offer

moderating successful punishment were due to trials in which proposers offered 1 Cent ($p < .05$). As such, FRN amplitudes indicating successful punishment (following a sad facial expression after rejection) are more positive in trials where only 1 Cent was offered, and participants reacted with a more positive compared to negative FRN amplitudes to the presentation of the offer.

The effect of FRN amplitudes following the offer moderating honored reward were due to trials in which proposers offered 5 Cent ($p < .01$). As such, FRN amplitudes indicating honored reward (following a smiling facial expression after acceptance) are more positive in trials where 5 Cent was offered, and participants reacted with a more positive compared to negative FRN amplitudes to the presentation of the offer. No other effects of the neural responses to the offer on the FRN following the facial expression were significant.

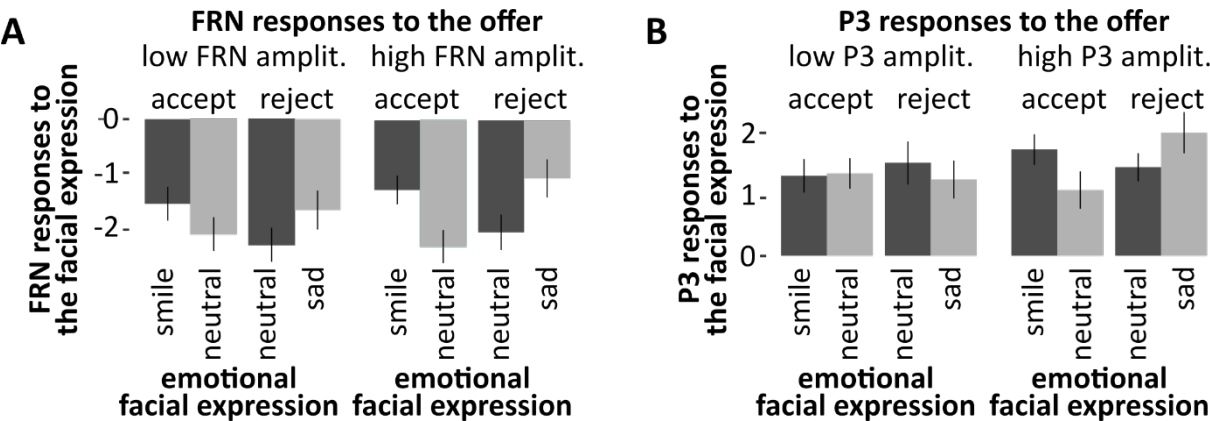


Figure S6: Results for moderating effects according to the neural response to the offer; error bars show the standard error around the mean. (A) Effect of emotional facial expression of the proposer (smile, neutral, sad) according to the decision of the responder and the FRN amplitude according to the presentation of the offer (depicted as low vs. high according to a median split) on N2 amplitudes following the presentation of the facial expression (in μV). (B) Effect of emotional facial expression of the proposer (smile, neutral, sad) according to the decision of the responder and the P3 amplitude according to the presentation of the offer (depicted as low vs. high according to a median split) on P3 amplitudes following the presentation of the facial expression (in μV).

Third, for the P3 following the facial expression, we found a moderating effect of the P3 following the offer on the interaction of decision and facial feedback (remember that the latter reflects potential effects of successful punishment and honored reward which was insignificant in the analysis reported in the main manuscript). In particular, the three-way interaction between decision, facial expression, and P3 amplitudes following the presentation of the offer on FRN amplitudes following the facial expression was $F = 14.5$, $p < .001$. As depicted in Figure S6B, an effect of successful punishment (more positive P3 amplitudes after sad compared to neutral facial expression following rejection) and honored reward (more positive P3 amplitudes following smiling compared to neutral facial expressions after acceptance) revealed if the P3 amplitudes following the offer were more (compared to less) positive. As a post hoc test, the interaction between decision and facial feedback on P3 amplitudes following the presentation of the facial expression was significant in trials in which participants (according to a median split) showed a stronger positivity to the presentation of the offer $F = 8.2$; $p < .01$.

Discussion

The analyses presented in this supplement are complex and were not pre-registered. So, it is important to stress the explorative character of these results. Starting on a more general level, the reported effects do provide evidence for the hypothesis suggested by an anonymous reviewer that expectations may have an influence on how the facial expressions were processed. Also, the structure of the results was stringent in that from the six analyses, in the two that showed effects, there was a contingency between the moderating ERP component and the dependent ERP component: FRN responses following the offer moderated FRN effects following the presentation of the facial expression, and P3 responses following the offer moderated P3 effects following the presentation of the facial expression. Contrary, there were no cross-effects (such as P3 moderating FRN responses), and no effects on N170. As a final more general note, in both cases we found a moderating effect on the interaction between decision and facial expression, i.e., the effect related to our main hypothesis of successful punishment and honored reward.

It is quite likely that many dynamic processes that take place across the trials of the experiment and across participants are not captured by the experimental conditions of our experiment. To illustrate, participants may show shifting states of attention during the course of the experiment due to fatigue; they may react differently to the emotional facial expressions according to their mood on that day, or they may build up specific expectations according to individual levels of fairness consideration or altruism or according to dynamic process resulting from specific sequences of trials. The analyses presented in this supplement cannot reveal the sources of these dynamics, yet, by investigating neural responses to the offer, provide an indicator of the states in which participants were during a trial before viewing the emotional facial expression.

Regarding the FRN following the presentation of the offer, we found a more pronounced effect of successful punishment and honored reward when FRN responses following the offer were more positive compared to negative. We performed follow-up analyses on this three-way interaction and, in subsequent analyses, also considered the size of the offer as additional moderating effect as the meaning of the FRN is more easily understandable if the valence of the eliciting stimulus is known. It turned out that the effect of honored reward was more pronounced if participants reacted with more positive compared to negative amplitudes to an offer of 5 Cents. As a potential, yet speculative post-hoc explanation, we suggest that the more positive FRN amplitude following the presentation of 5 Cents might indicate that participants were particularly delighted for receiving the maximum possible offer, a fair split of the pot. The positive emotional state is amplified during the positive interaction, particularly a smile for accepting the offer. Transferring this scenario to an in-person-interaction, we might consider an individual being particularly delighted for receiving a fair offer to smile when accepting the offer; if the partner smiles back, the positive interaction elicits a pronounced positive response in the receiver, reflected in more positive FRN amplitudes following the smile. A different interaction between prior states of the receiver and emotional feedback of the proposer would occur when the offer of 5 Cents would be perceived as less positive (for example, because the participant is fed up for having received low offers in the past, or is in a general bad

mood, and perceives the offer as "the very least"). Even a smile after acceptance would not be perceived as overly positive.

The effect of successful punishment on the FRN (FRN amplitude following sad facial expression after rejection) was moderated by FRN-related neural responses to the presentation of an offer of 1 Cent. We speculate that in trials in which participants reacted particularly negative (e.g., angry) to an offer of 1 Cent, the sad facial expression, which might imply an apology-like signal, is ignored. Relatedly, the anger about the unfair offer might be so strong as it conceals the meaning of the sad face in terms of an understanding, apologizing, and affirming to change in the future. Contrary, more positive FRN amplitudes might, despite the fact that only 1 Cent was offered, indicate that participants are still engaged in the interaction process and eager to influence the proposer in order to make better offers in the future. Accordingly, the sad facial expression signaling that the message was received is particularly rewarding.

Regarding the P3 effect following the presentation of the offer, we might speculate that individuals with high compared to low amplitudes are currently more engaged. P3 amplitudes are associated with the perceived salience and reflect the motivational significance of a stimulus. As such, offers which elicit a higher compared to lower P3 amplitude might be perceived as being more important or relevant and might indicate trials in which individuals are more engaged. Thus, in such trials, participants might have more pronounced expectations regarding the emotional facial expressions, and these stimuli might be considered more carefully, processed more deeply, or observed more attentively. Accordingly, any effect that these stimuli produce should be more pronounced. This is in line with the moderating effect we found, i.e., the expected effect for successful punishment and honored reward in those trials in which participants reacted with a stronger P3 amplitude towards the offer.

We re-iterate that these interpretations of the exploratory post-hoc analyses are very speculative.

Yet, they at least suggest that interesting dynamic processes are going on beyond what we initially

293 intended to investigate, and that these dynamics are worth for in-depth exploring and open up
294 fascinating new research opportunities.

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